ARM Graphics

Ed Plowman

Graphics Product Marketing Manager

Growing & diverse need for embedded graphics

Safety Critical

Avionics

Automotive

Consoles

Settop Boxes

Common Goals:

Efficiency

Image Quality

Performance

Phones

PDAs

PMPs

The design goal is efficiency in area, power consumption and bandwidth utilisation without compromise to image quality features or performance.

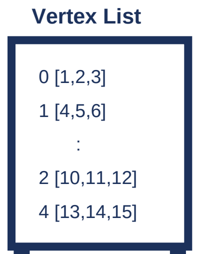
ARM is building a range of graphics processing units (GPU's) to meet the requirements of the varying key embedded markets meeting these design goals with a vibrant eco-system to match.

ARM Mali Graphics Cores Overview

- Efficient Memory Bandwidth Usage
  - Proprietary combined tile-based & immediate mode rendering
  - Smart data-structures and flow algorithms
- High Performance and Image Quality
  - 4X / 16X Full Scene Anti-Aliasing – penalty free 4X
- Rich Feature Set for Differentiation
  - 3rd generation IP cores takes 3D graphics beyond Shader Model 3.0
  - Architecture allows for other general multimedia acceleration
- Low Production and Integration Cost
  - Acceleration beyond 3D graphics (OpenVG, etc.)
  - Market leading PPA, solutions from 1.1 - 5 mm^2 in 90nm process
  - Internally developed and pre-verified software stack
- Upwards and downwards scalable solutions
  - Multi-core
  - Same software stack

- **Zeppelin Demo**
- **OpenGL ES 1.1 common lite**
- **Antialiasing (16xFSAA)**
- **Texture compression**
- **Multitexturing w/mipmapping**
- **Trilinear texture filtering**
- **Alpha blending**
- **Reflections**

Traditional Vs Tile Based Rendering



Traditional (Immediate mode) Rendering:

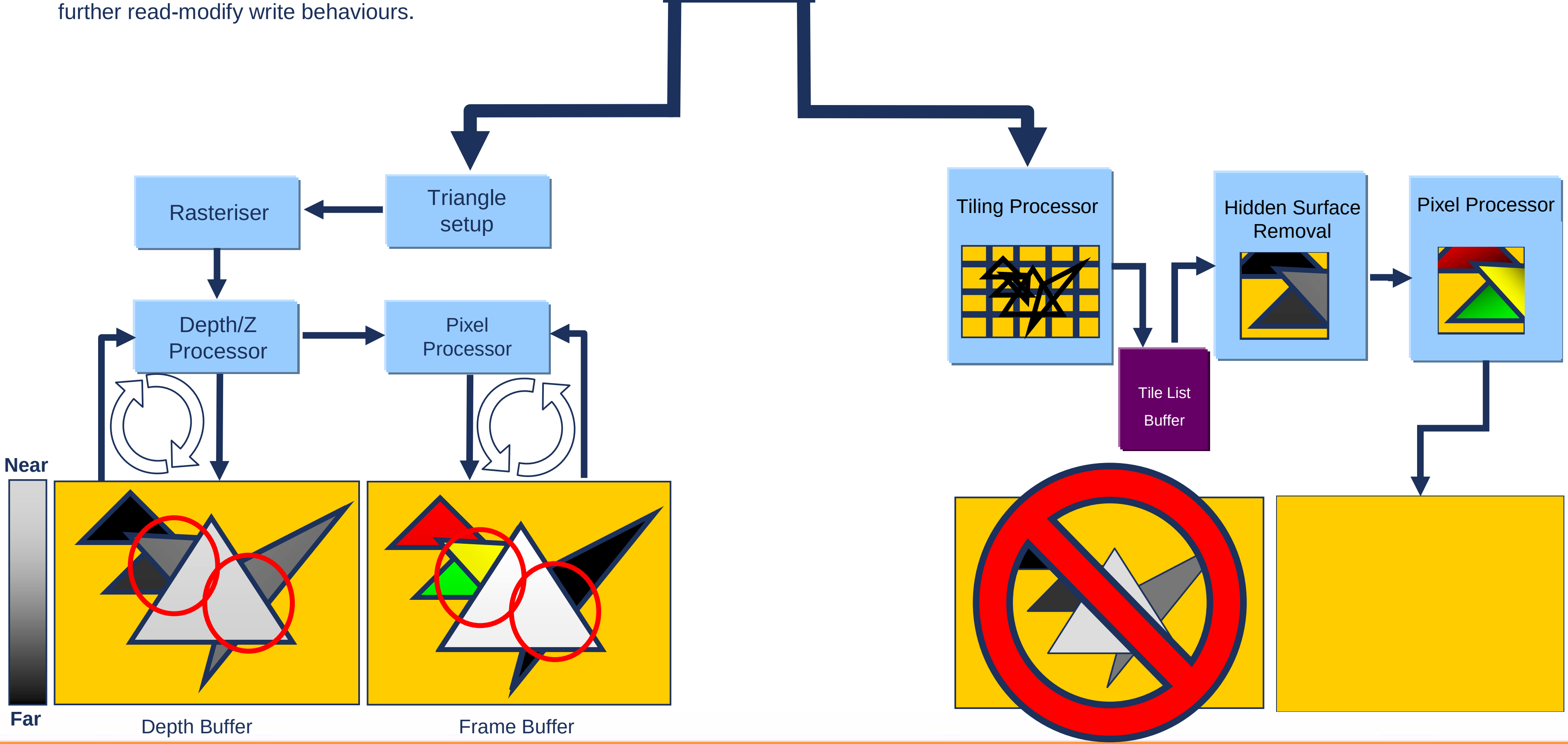
Pros: Simple Architecture, Hi through put, Linear scaling performance with scene complexity.

Cons: Hi B/W usage due to read-modify-write behaviour, extra memory foot print for z-buffer, using alpha creates further read-modify write behaviours.

Tile Based (AKA Deferred) Rendering:

Pros: No Z-buffer, lower bandwidth, no redundant processing or memory accesses.

Cons: Complex architecture, hi scene complexity can lower efficiency.



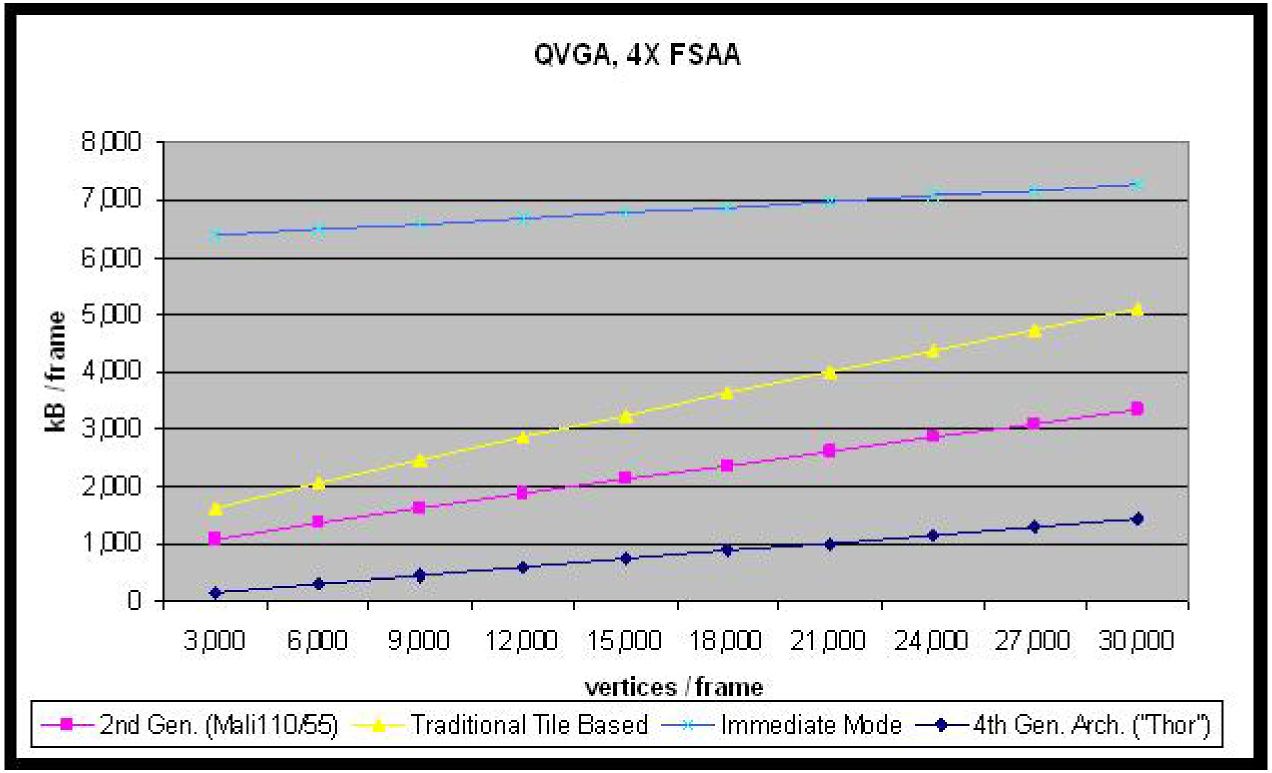
Mali – hybrid rendering approach

- The best of both worlds…
  - Compared to traditional immediate mode renderer
    - 80% lower per pixel bandwidth usage
      - Even with 4X FSAA enabled
    - Efficient memory access patterns and data locality
      - Enables performance even in high-latency systems
  - Compared to traditional tile-based renderer
    - Significantly lower per vertex bandwidth
    - Significantly reduction in impact of scene complexity
    - Simple and high-performance HW / SW interface
  - Architectural advantages
    - Per frame autonomous rendering
    - No renderer state change performance penalty
    - On-chip z / stencil / color buffers – minimize working mem. footprint

- HauntedHouse - OpenGL ES 1.1:
- Antialiasing (16xFSAA)
- Texture compression (FLXTC)
- Multitexturing w/mipmapping
- Trilinear texture filtering
- Environment mapping

The Mali Advantage

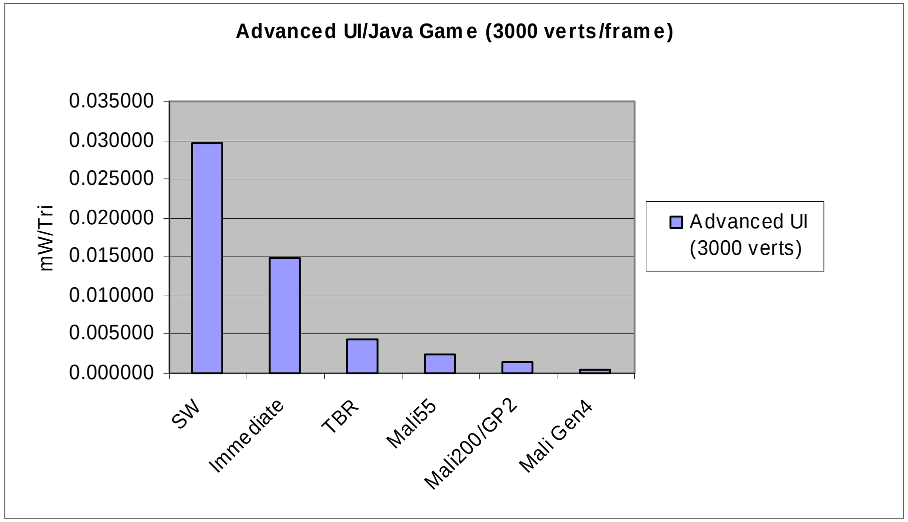
Bandwidth Comparisons of existing architectures Vs Mali hybrid architectures

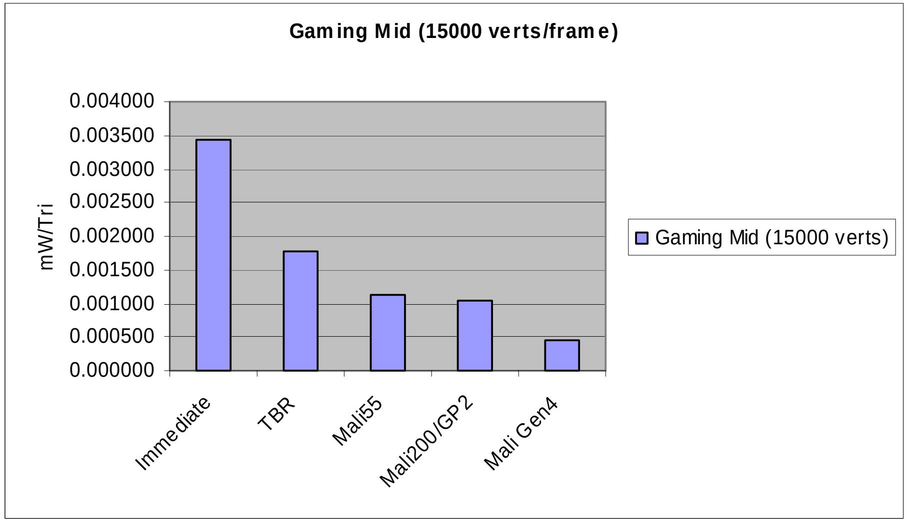


The Mali architecture already offers better bandwidth efficiency than its rivals with further enhancements planned for the 4th generation core.

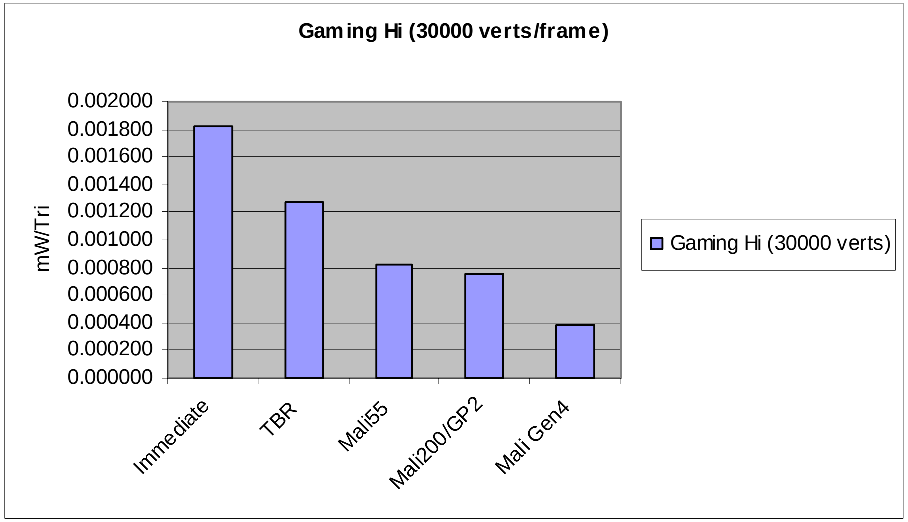
Calculated using Molnar-Eyles formula (triangles evenly distributed filling the frame-buffer size in all layers (overdraw+1), assuming an overdrawn factor of 4.

Power efficient graphics using Mali





- The graphs show an estimate of the total mW cost for three use cases.
- Power calculations take into account core and memory system power consumption for each architecture.
- Access to external memory = 10x more mW than on chip activity.
- Mali's bandwidth saving gives a clear power advantage Vs the competition.



Mali Product Range

Mali™ 2007 Product Line-up

- **Mali55**
  - Worlds smallest GPU
  - Low-end devices – bringing HW acceleration to where it was not possible before
- **Mali200 / MaliGP2**
  - Fully programmable “desktop” featured GPU for WVGA and beyond
  - Programmable Graphics Core for the convergence and high-end markets

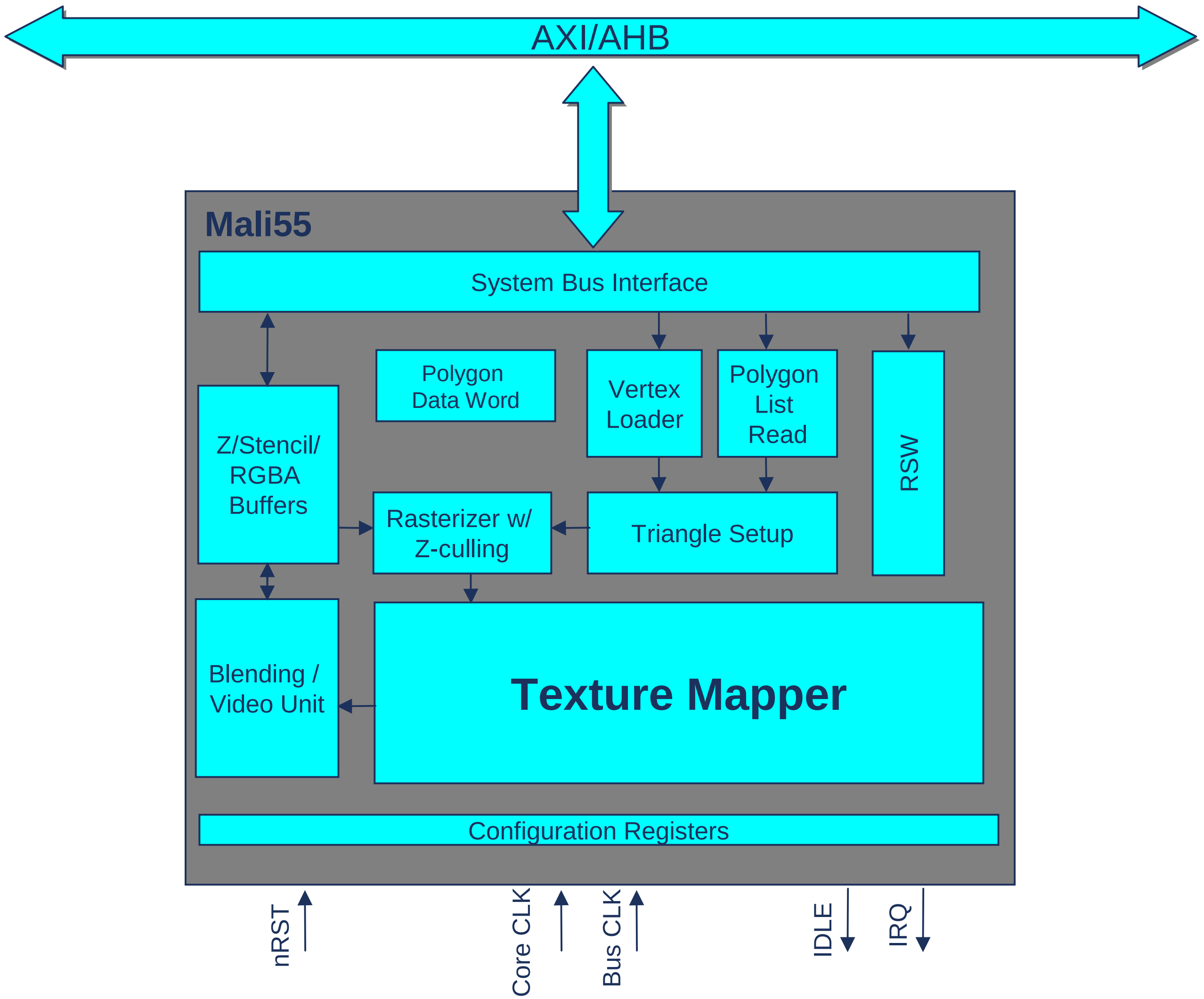
Mali55 Core - Features Detail

Key Features	System features
▪ OpenGL ES v1.1 Feature Set with extensions	▪ 16 / 32 bit frame buffer
▪ 4X/16x FSAA	▪ Max. Resolution 2048x2048
▪ 2bpp Texture Compression (FLXTC) + ETC	▪ Autonomous Frame Rendering
▪ On-chip Z / Color / Stencil Buffers	▪ Memory Management Unit

Accelrerated Hardware Feature Highlights

- Points / Lines / Triangles / Quads
- Flat / Gouraud Shading
- Perspective Bi- / Tri- linear Filtering
- Multi texturing / RTT
- Auto Mip Map Generation
- Dot3 Bump Mapping
- Flexible Texture Input formats
- Specular Color / Color Sum
- Render To Texture w/ AA
- 2D / Point / JSR184 Sprites
- Bitblt / ROP3/4
- OpenVG Blending, Scissoring, Masking, Paint Generation, Image Interpolation

Mali55 Block Diagram



Mali200 & GP2 Core Features

Basic 3D Features

- Perspective Bi- / Tri- / Anisotropic Filtering
- 4X and 16X FSAA (4X penalty free)
- 4-level hierarchical Z and Stencil Tests
- Frame buffer blend with Destination Alpha
- Scalable performance with Multi-cores

Pixel Shader Model 3 Features

- Cube / 3D / Projective / Shadow textures
- Unlimited dependent texture reads
- Programmable LOD bias
- Multiple Render Targets
- Two-sided Stencil
- Fast dynamic branching
- Full Floating Point Arithmetic
- HDR textures and frame-buffers
- Complete non-power-of-2 texture support

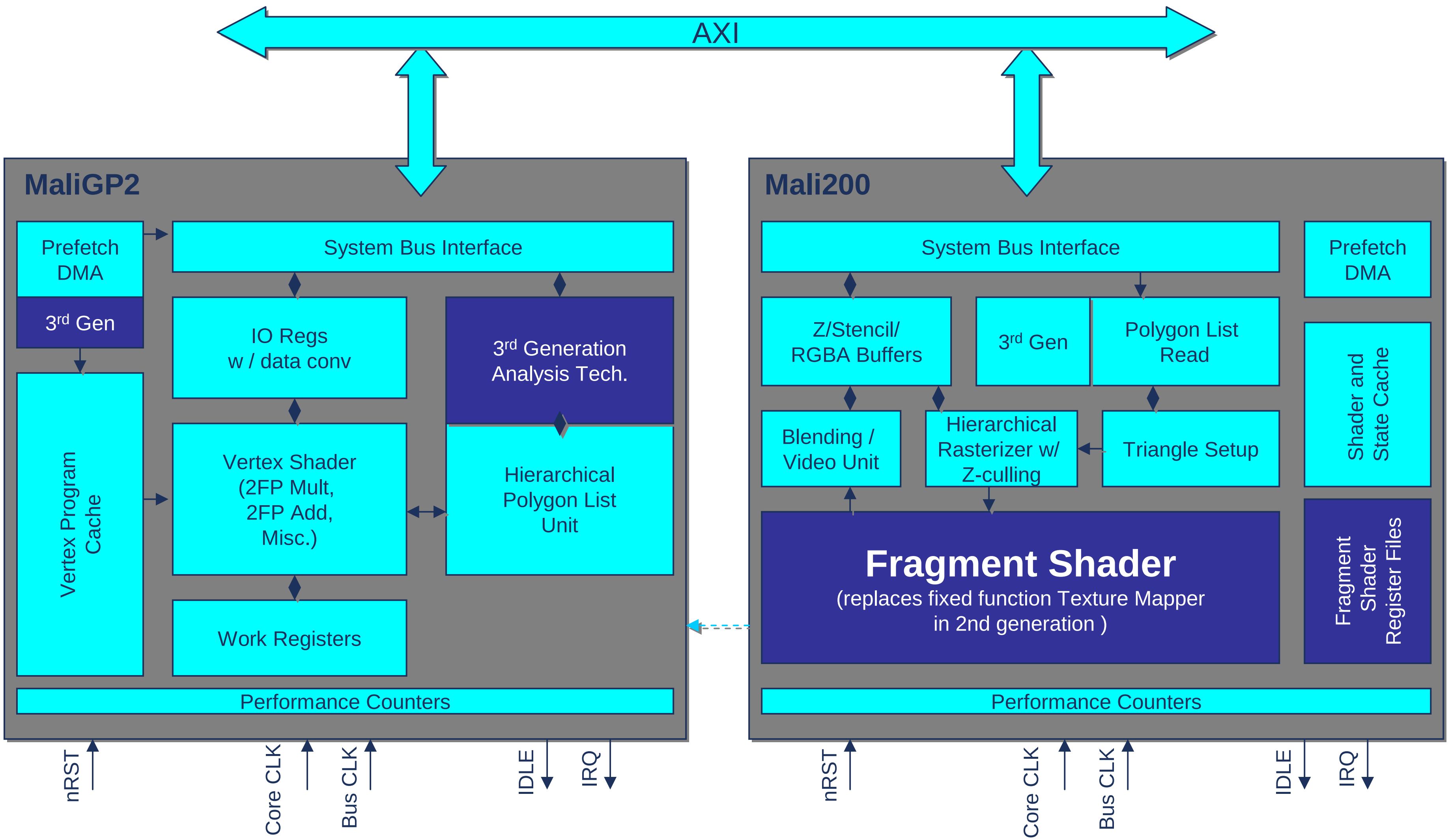
Beyond Shader Model 3 Features

- No program length limits
- Virtualized texture samplers (no texture count limit)
- Indexable varying, temp and texture samplers
- Fast Arctangent
- Dynamic Recursion
- Shader access to frame-buffers
- Register Indirect Jumps
- Arbitrary Memory Reads and Writes
- Microsoft Vista Ready

Other Multimedia Feature Highlights

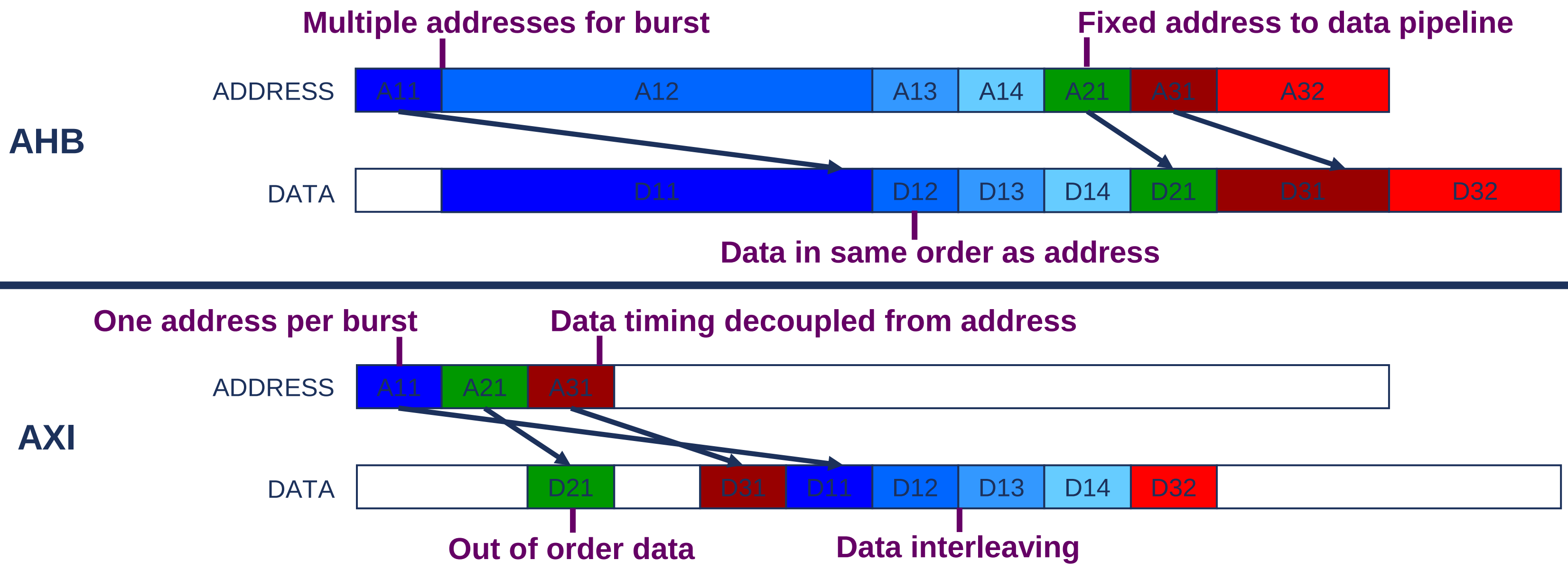
- 2D Primitives (Lines, Squares, Triangles, Points)
- ROP3/4
- Arbitrary Rotation / Scaling
- Alpha Blending
- Multitexture BitBLT
- OpenVG / SVG Acceleration

Mali200 & MaliGP2 Block Diagram



Mali supports AXI

- ARM Graphics solutions AXI ready!
- Easier integration with new AXI ready cores and fabric IP
- Greater bus efficiency
  - Transaction model improves access to available bandwidth
  - Mali core tuned to AXI transaction model, less latency sensitive
  - Fewer busses and MHz required to support a given system bandwidth



Mali Product Fundamentals

	Mali200	MaliGP2	Mali55
Anti-Aliasing	4X / 16X	4X / 16X	4X / 16X
OpenGL®ES 1.x	YES	YES	YES
OpenGL®ES 2.x	YES	YES	NO
OpenVG 1.x	YES	NA	YES
Gate count Logic	1275k	280k	320k
SRAM	35kB	13kB	5kB
Die Area 90nm (util. 80%)	5.5mm^2	1.5mm^2	1.4 mm^2
Power 90nm	0.7mW/MHz	0.3mW/MHz	0.22 mW/MHz
Max CLK	275MHz	275MHz	200MHz
Fill rate / s	275 Mpix	NA	100 Mpix
Triangles / s	9M	9M	1M
DirectX9 Vista ext.	YES	YES	NO

Rendering and Computation Equivalence Performance Notes

- Performance figures based on Max CLK
- 4X FSAA turned on (1 pix = 4 sub-samples) and bi-linear filtering
- Fill rate and geometry expressed as max theoretical performance

Area, Gate Count, frequency Notes

- Advantage CL90G 125C, 0.9V, slow silicon, Full Scan, Nand2x1AD cell size: 2.8224um^2
- Overweight of Single-Port SRAM
- Max frequency

Mali Supports OpenGL ES

- Small-footprint subset of OpenGL
  - Created with the blessing and cooperation of the OpenGL ARB
  - Eliminates redundancy and desktop centric functionality
- Powerful, low-level API with full functionality
- Now the de facto standard for embedded graphics
  - Available on all Mali cores

Java MIDP1.0

Native SW app

OpenGL ES Versions

- ES 1.0
  - Fixed-function pipeline - Optimized for SW renderers (ARM9 and up)
  - First implementation shipped in product late 2004 (Nokia 6630)
- ES 1.1
  - Fixed-function pipeline– Targeted more towards hardware acceleration
  - First implementations started shipping in 2005
  - Will continue to be sweet spot for next 18 months in product
- ES 2.0
  - The next generation – programmable graphics pipe using shaders
  - Shaders written in high-level “C-like” language and compiled for hardware
  - Offers greater flexibility and potential efficiency for high image quality than ES 1.1
  - Not backwards compatible, but can implement ES 1.1 driver on 2.0 hardware

OpenGL ES 2.0: Whats the big deal?

- OpenGL V2.0 & OpenGL ES V2.0 major feature is programmability through shaders

- Current state of the art in desktop
  - New games starting to use Shaders
  - Next step in visual quality

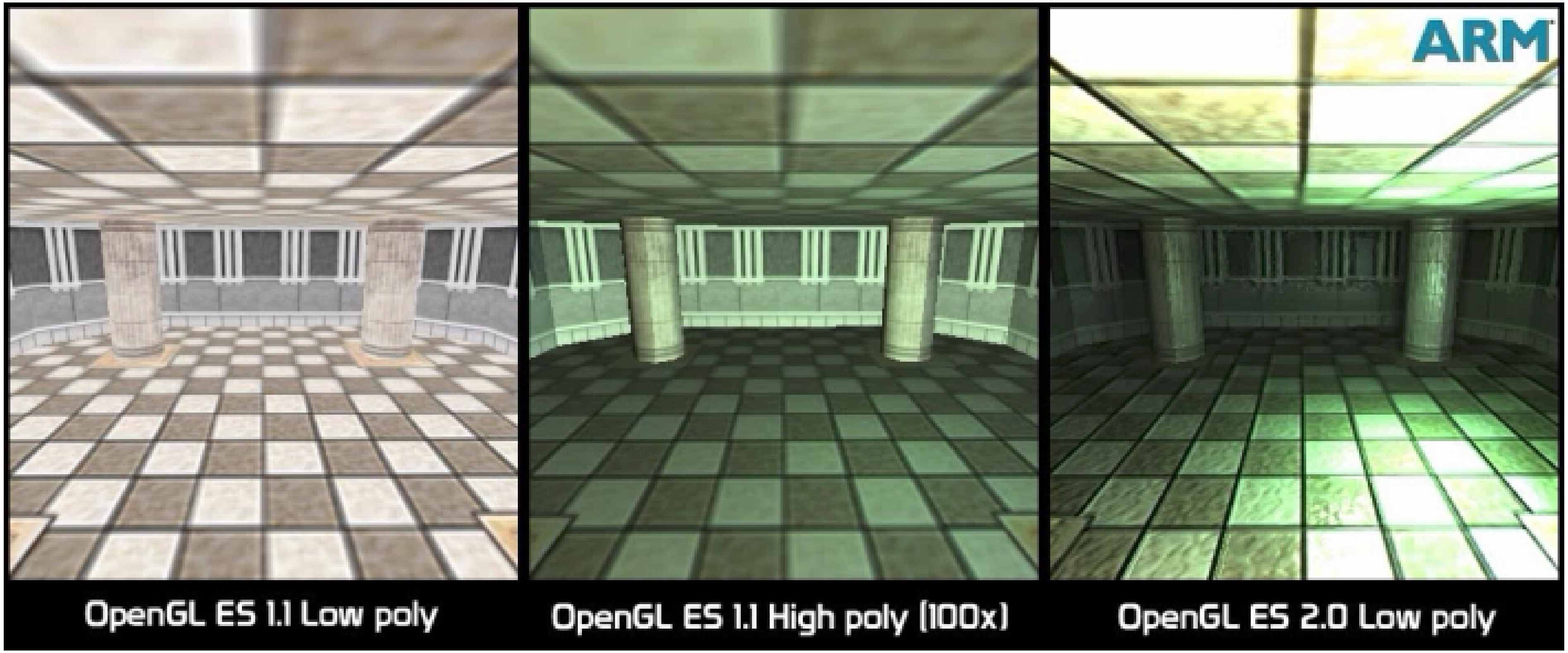
- Sounds “Hi End” is there need in handheld?
  - Physical limitation on screen size = Fixed DPI
  - In turn mean little advantage in high polys/sec figures
  - Pushes avg poly area lower and lower
  - ROI diminishes sharply Vs processing required

- Shift in metric required
  - Polys/sec no good, need to look at vertex & pixel ops/sec.
  - If poly processing overhead > contribution to visible scene
  - Better spent on complex vertex and pixel level manipulation
  - This is the basic principle behind

- **Mali200 family targeted at full OpenGL ES V2.0 support**

The shader advantage: Quick Example

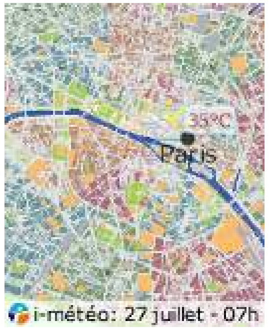
- Objects in Images generated from the same basic data set

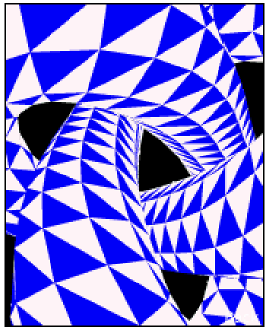


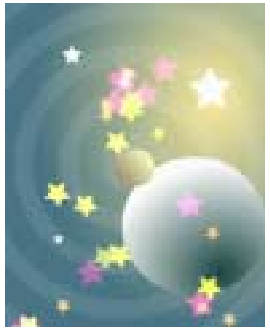
- Ability to manipulate basic data at vertex and pixel level through shader programs leads to better image synthesis without extra input data

Mali supports OpenVG

- Many applications use low-level vector graphics
  - Cartography/GPS applications, E-book Readers, etc.
  - Advanced user interfaces
- Many vector graphics formats in use
  - Flash, SVG, PDF, Postscript, Vector fonts etc. etc.
- OpenVG gives access to acceleration of existing formats
  - NOT a competitor to existing formats

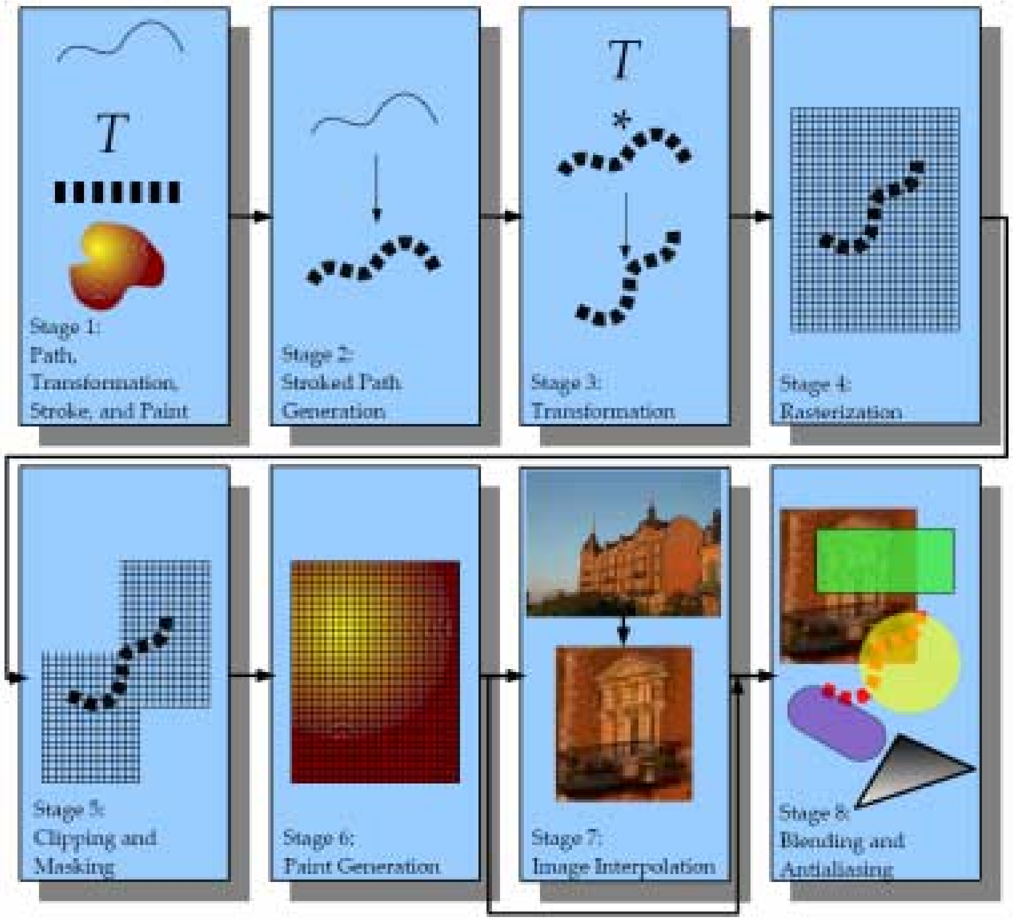






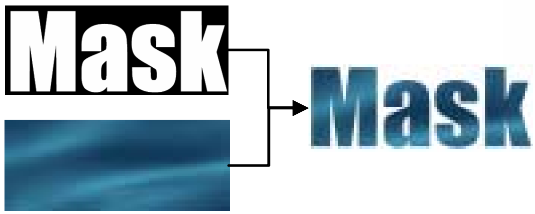
OpenVG Pipelines

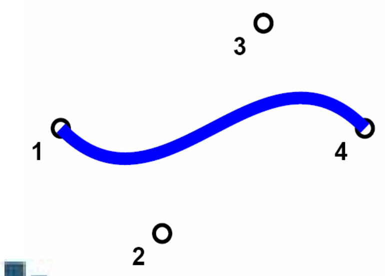
- Definition of path, transformation, stroke and paint
- Stroked path generation
- Transformation
- Rasterization
- Clipping and Masking
- Paint Generation
- Image Interpolation
- Blending

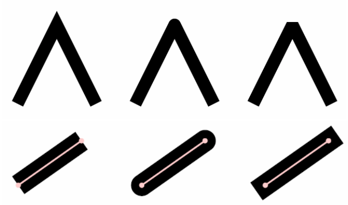


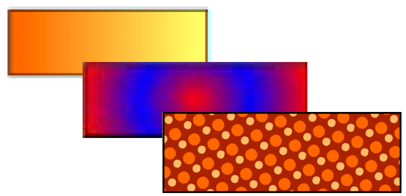
OpenVG Features Support In Mali

Functionality	Mali55	Mali200
Blending (including DARKEN and LIGHTEN)	Y	Y
Coverage	Y	Y
Scissoring	Y	Y
Masking	Y	Y
Paint generation	Y	Y
Image interpolation and drawing modes (including STENCIL mode)	Y	Y
Image Filters (Gauss blur, colour matrix, convolution etc.)	N	Y

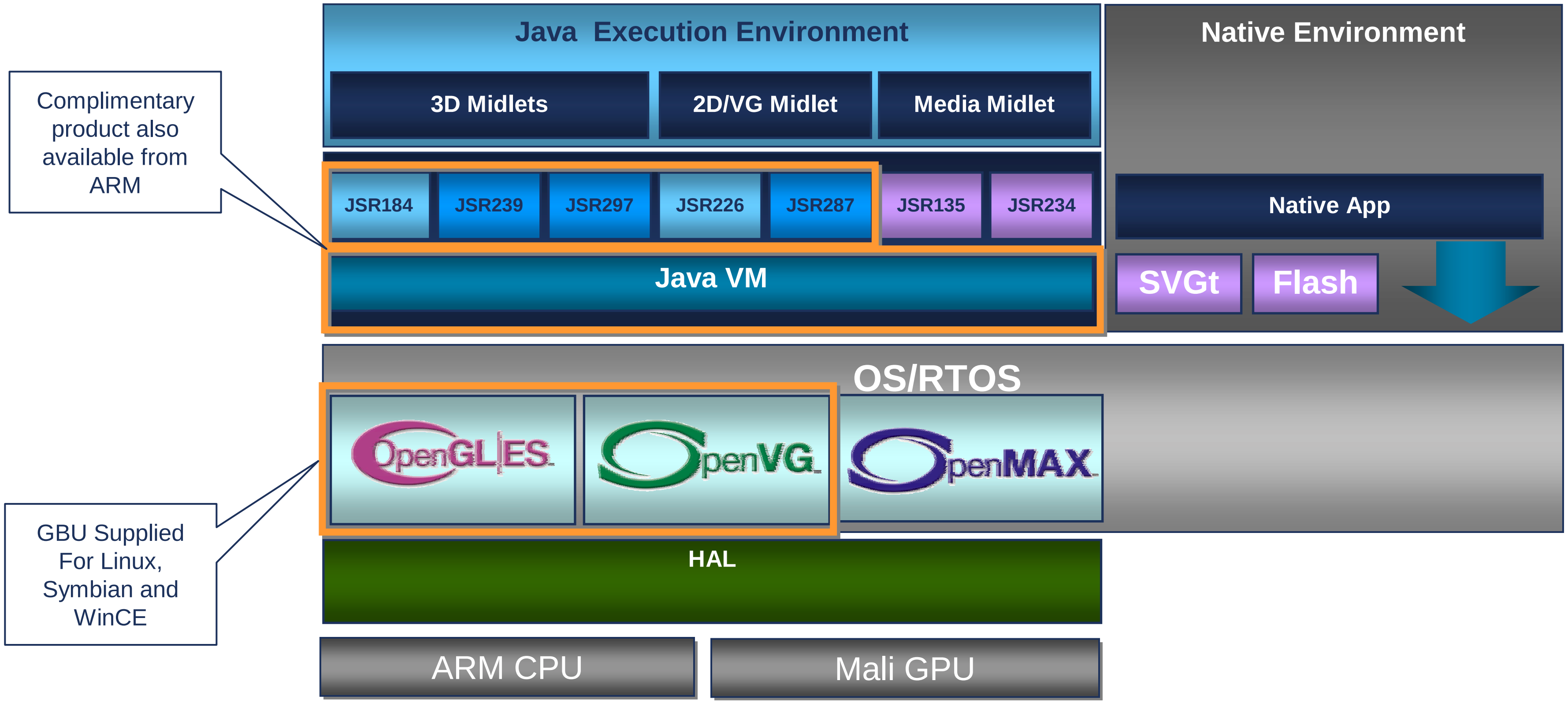








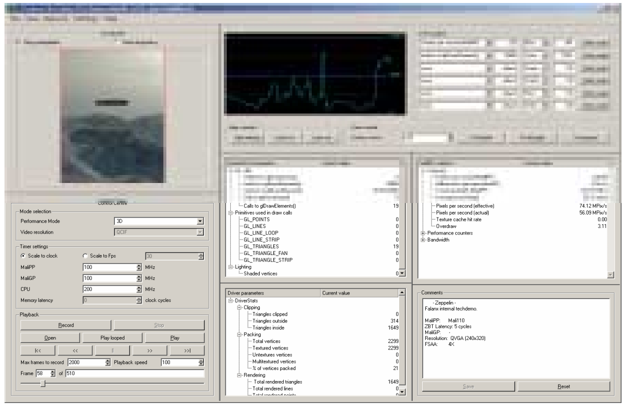
Mali Software Support



- Complementary middleware components developed by ARM
- Used in conjunction with third party IP to validate the Mali hardware solution (and associated drivers)
- Commercial grade middle products offered to OEMs for use in handsets
- Middleware tuned to deliver optimum performance with Mali product family

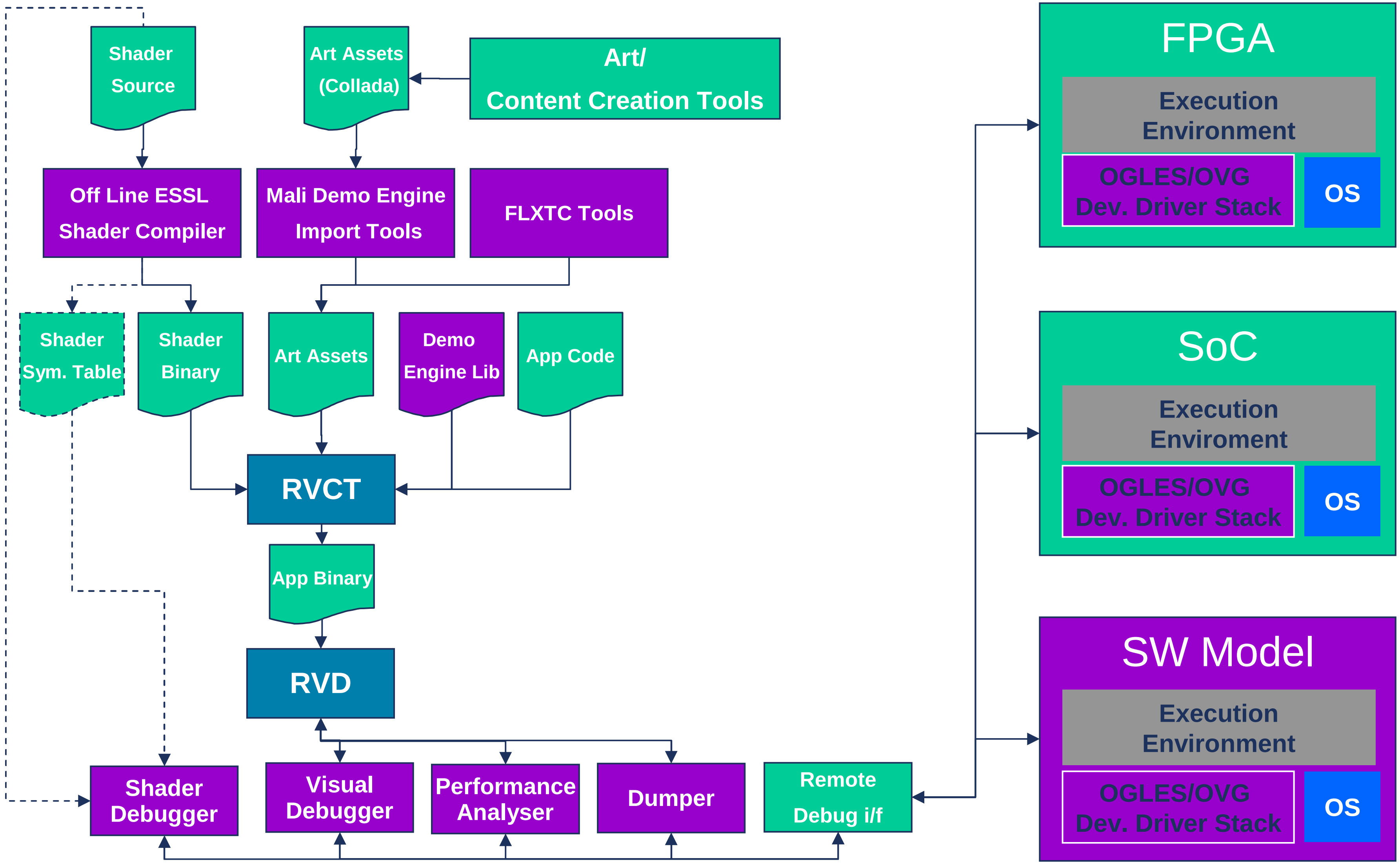
Software package

- Supporting Development Tools
  - Performance Analysis Tool – (PAT)
  - Texture compression (FLXTC)
  - Technology Demos
  - Visual Debugger
  - Shader Debug Tools



- Software documentation
  - Application developer guide w/ performance analysis considerations
  - Software Reference Manual
  - Software Integration Manual (OS porting guidelines)
  - Tools documentation (PAT, FLXTC, Shader Debug and Compiler)
- Application test suite
  - Comprehensive Test suite for all API drivers
  - Test description and coverage analysis
  - Support for Khronos Conformance Testing (OpenGL ES, OpenVG)

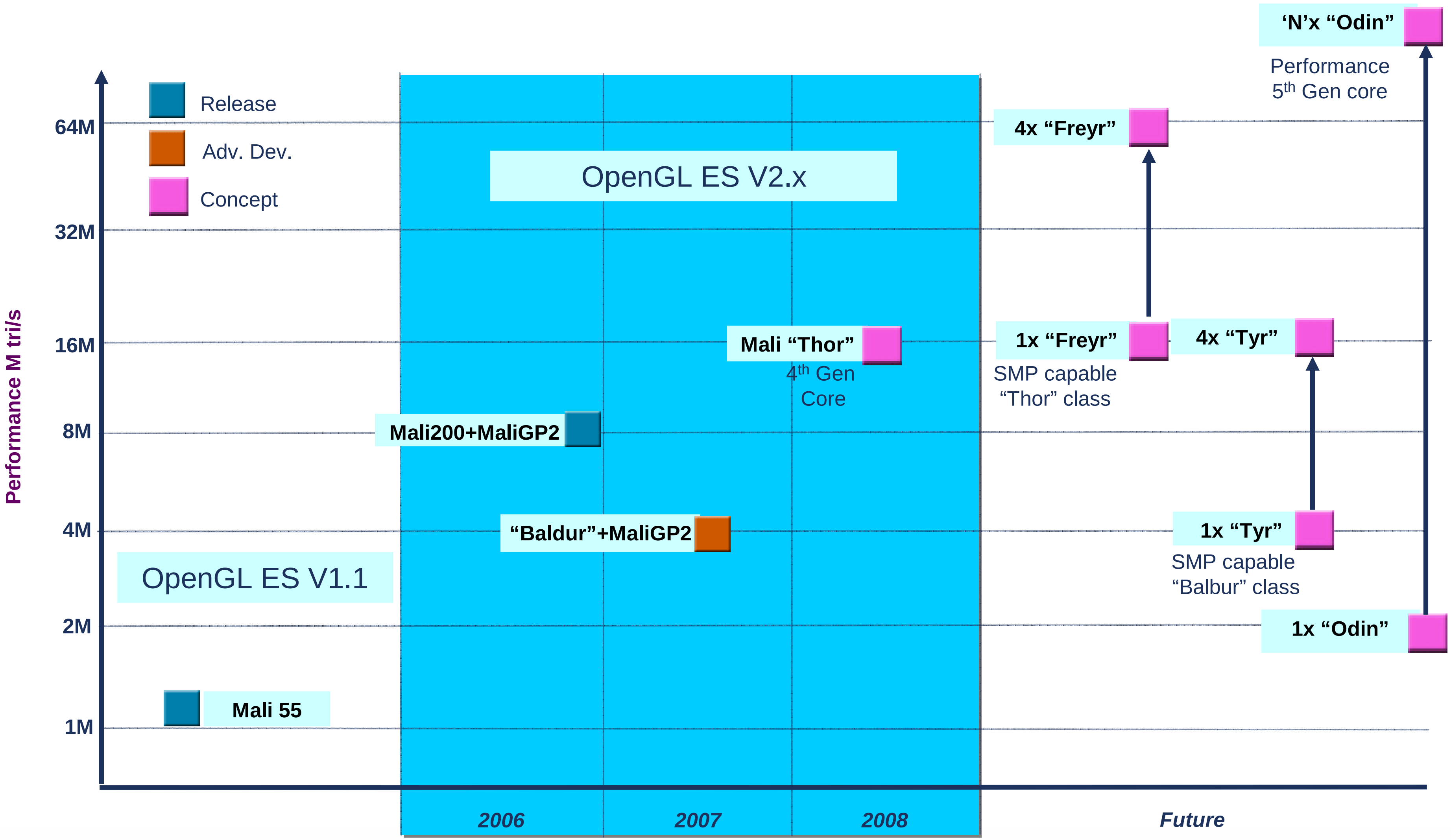
ARM Graphics Development Tools



= SDK Component
= ARM Development System Tools

Mali Roadmap

Graphics Roadmap



Baldur

- Goal is 50% size with ≤ 50% performance drop Vs Mali200
  - New IP to reduce SRAM instances, sizes and logic
  - Algorithmic redesign of caches and cache tags
  - Some optional components in config are being considered for PPA trade offs

- Binary compatible with Mali200 and MaliGP2

- Possibilities of Multi-core and optimized L2-cache for lower bandwidth usage

Thor

- Architectural changes to both Mali200 and MaliGP2
- Performance goal: 2X Mali200/GP2
- Size goal: <50% increase on Mali200/GP2
- Memory bandwidth: 50% decrease, not affected by tiling
- Algorithmic improvements for MRT, Pixel and Vertex Shaders
- D3D10 feature set – geometry shaders
- Building Multi Processing GPU Capability
  - linearly scalable performance per core
  - Increased efficiency through coherency
  - Exploiting AMBA 4.0 feature set
- New low-level software drivers to accommodate

END